ECE315 Introductory Microprocessor Laboratory

Lab 1

Schematic Symbols, PCB Footprints, and Defining Board Shape

1. Introduction

An embedded system is a collection of integrated circuits, sensors, and other components that are chosen to accomplish a specific task. These components are soldered onto a printed circuit board that interconnects the components as specified by the system designer.

You have all probably taken other laboratory courses where you design a circuit using a breadboard and through hole parts. This is a great way for students to learn key concepts and take measurements used to validate their designs. Industrial design of embedded systems is much different. Instead of using bread boards, we use a printed circuit board (PCB) to help ensure reliability and manufacturability that cannot be accomplished with a breadboard. This course will introduce you to some of the tools and practices used to design a PCB.

When designing an embedded system, an embedded systems engineer will use a schematic editor to generate a set of schematics for a design. The schematics define how each component is used and how components are interconnected.

This semester, you will design a Whack-A-Mole game using a Raspberry Pi Pico microcontroller and various other components. This design will be fairly straightforward, but it will help you to learn some of the basic principles of schematic design and PCB layout.

2. Altium Designer

ECE315 will require you to use Altium Designer to complete a custom printed circuit board (PCB). Altium Designer is available on any CAE PC. You will be required to create an AltiumLive account in order to use Altium on the CAE PCs. Once you have created an account, you will also be able to download and install Altium on your own PC (Altium is a Windows Only application).

The instructions below will include commands that can be issued in Altium. The commands will be highlighted in **bold red letters**.

3. Base Project Download

Download the Altium base project from the Canvas Lab 1 assignment page. You should extract the contents of the ZIP file to your CAE network drive. You will use this base project for labs 1-3.

4. Altium Settings

Use the following link to configure Altium Designer. These settings in this document will help during the design process.

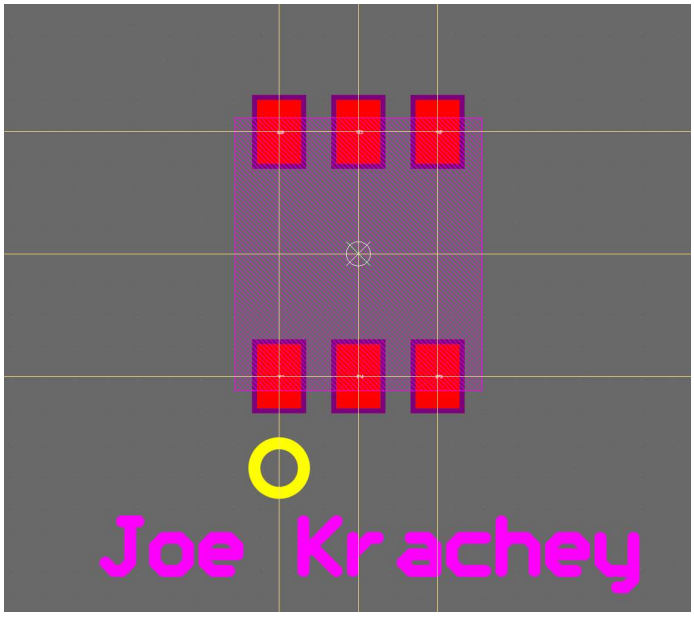
5. PCB Footprints (T3A33BRG-H9C0002X1U1930)

An embedded system is a collection of individual components that are interconnected on a printed circuit board. Before you can begin to design your system, you will need to create several components using Altium Designer. The first of these components will be an LED with an embedded serial interface that allows an MCU to control the color displayed on that LED.

Use the information provided in this document to create the PCB Footprint for this part.

Submission Requirement

1. Complete your PCB Footprint for the T3A33BRG LED
2. Add your name on Mechanical 1 just under the footprint by typing 'ps'
3. Use Snipping Tool to create an image of your footprint.
4. Save the file as Footprint-T3A33BRG.png
5. Upload the file to Lab 1 Assignment in Canvas.



6. PCB Footprints (CMT-8540S-SMT)

You will need to create the PCB Footprint for a magnetic buzzer that will be used in your design to generate tones/sounds. The process will be very similar to what you have completed in the previous section.

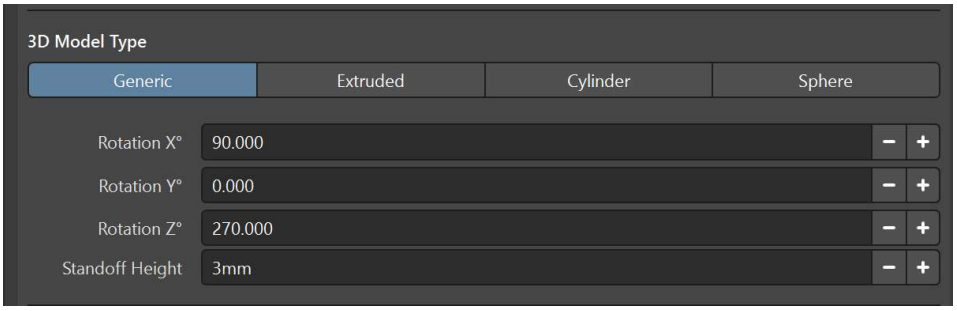
Footprint Requirements

1. When selecting the reference designators, use P and N for the positive and negative pads of the buzzer. The other two are used strictly for mechanical support. You can name these pads M1 and M2.

2. Add a '+' sign on the Top Overlay near the P pad (**ps**). This will be used during assembly to determine how to orient the part.

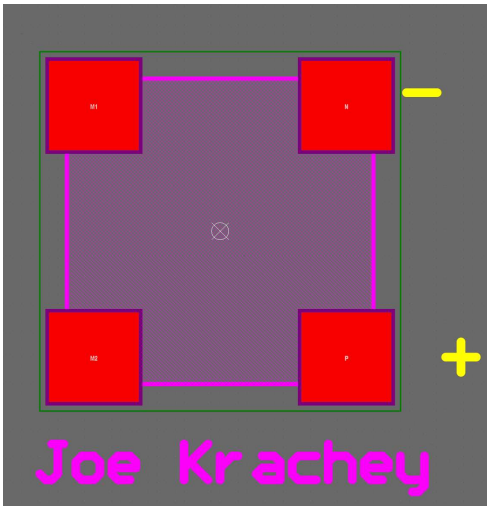
3. Add a '-' sign on the Top Overlay near the N pad (**ps**). This will be used during assembly to determine how to orient the part.

4. Make sure to add the 3D body (./3D/CUI_DEVICES_CMT-8540S-SMT-TR.step) to the footprint. You will need to rotate the 3D Body to get it aligned properly.



Submission Requirement

1. Complete your PCB Footprint for the CMT-8540S-SMT
2. Add your name on Mechanical 1 just under the footprint by typing 'ps'
3. Use Snipping Tool to create an image of your footprint.
4. Save the file as Footprint-CMT-8540S-SMT.png
5. Upload the file to Lab 1 Assignment in Canvas.


M1
M
M2
P
Joe Krachey

7. PCB Footprints (MCP1316T-29LE/OT)

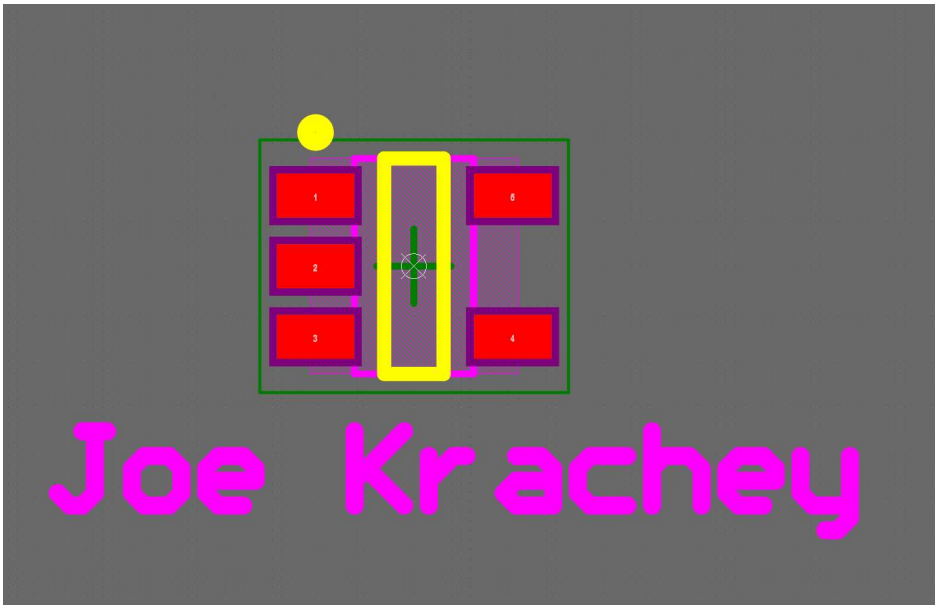
The final part you will create is a reset supervisor that is used to place the MCU in reset when the user presses a reset button OR if the 3.3V supply voltage were to drop below 2.9V.

The previous two parts required you to manually create the PCB footprint. This can be a slow and error filled process. Component manufacturers typically follow IPC standards that standardize assembly and production requirements of electronic equipment and assemblies. These standards help designers to ensure that the PCB Assembly companies will be able to successfully place components on a PCB.

Altium Designer supports an IPC Footprint Wizard that generates footprints for components. Use the following document to create an IPC compliant footprint for the reset supervisor.

Submission Requirement

1. Complete your PCB Footprint for the MCP1316T-29LE/OT
2. Add your name on Mechanical 1 just under the footprint by typing 'ps'
3. Use Snipping Tool to create an image of your footprint.
4. Save the file as Footprint-MCP1316T-29.png
5. Upload the file to Lab 1 Assignment in Canvas.



8. Schematic Symbols (T3A33BRG-H9C0002X1U1930)

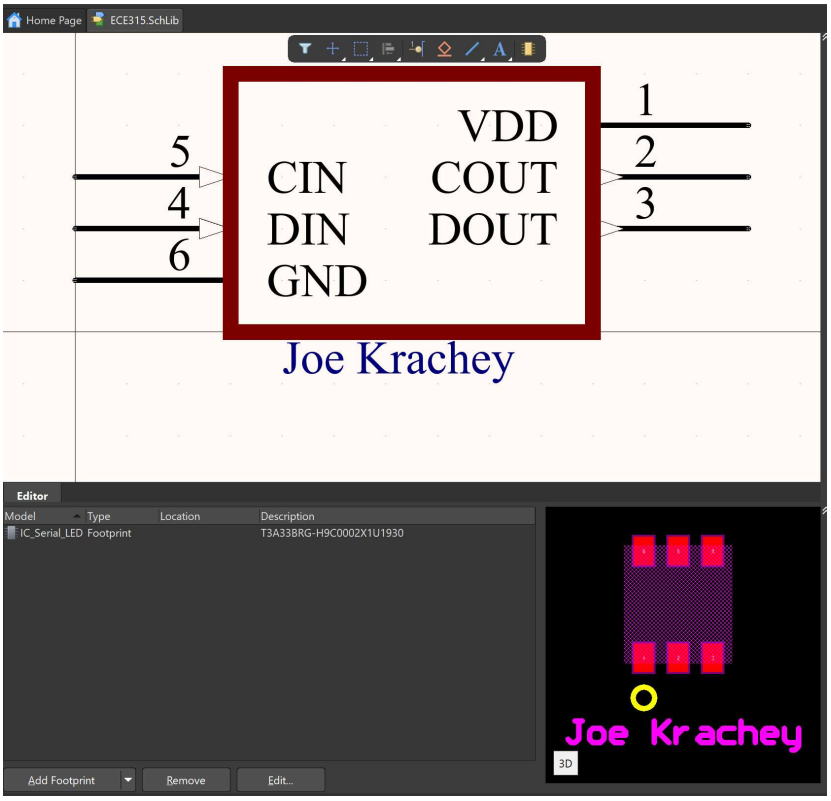
Now that you have your PCB footprints finished, you will need to create some schematics symbols for those components.

You can use the following document to create a schematic symbol for the T3A33BRG serial LED. Note: **You do NOT need to create a multi-part symbol for this lab, so you can skip section 7 of the document.**

Submission Requirement

1. Complete your Schematic Symbol for the T3A33BRG LED
2. Use Snipping Tool to create an image of symbol along with the preview of the footprint
3. Save the file as Symbol-T3A33BRG.png
4. Upload the file to Lab 1 Assignment in Canvas.

Make sure the pin numbers match the image below or the LEDs will not function correctly!



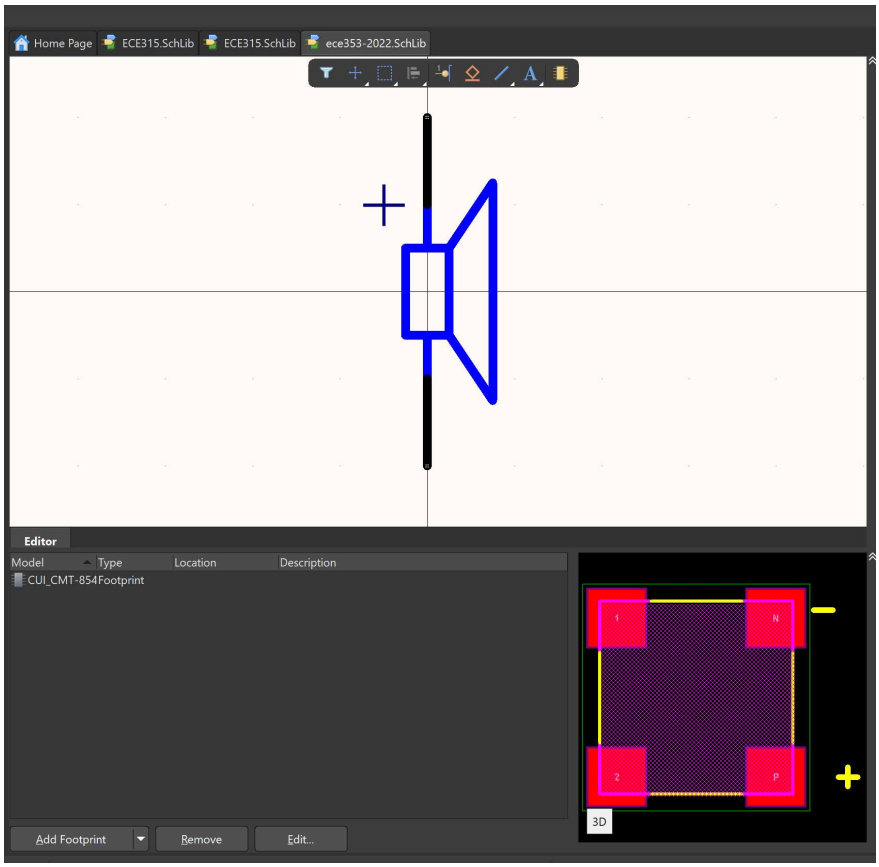
9. Schematic Symbols (CMT-8540S-SMT)

The schematic symbol for the CMT-8540S-SMT has already been created for you. You will need to add the footprint you designed to this symbol.

Submission Requirement

1. Associate the PCB footprint to the symbol.
2. Use Snipping Tool to create an image of symbol along with the preview of the footprint
3. Save the file as Symbol-CMT-8540S-SMT.png
4. Upload the file to Lab 1 Assignment in Canvas.

NOTE: The names are missing in the image below, but your screen capture should have your name visible!

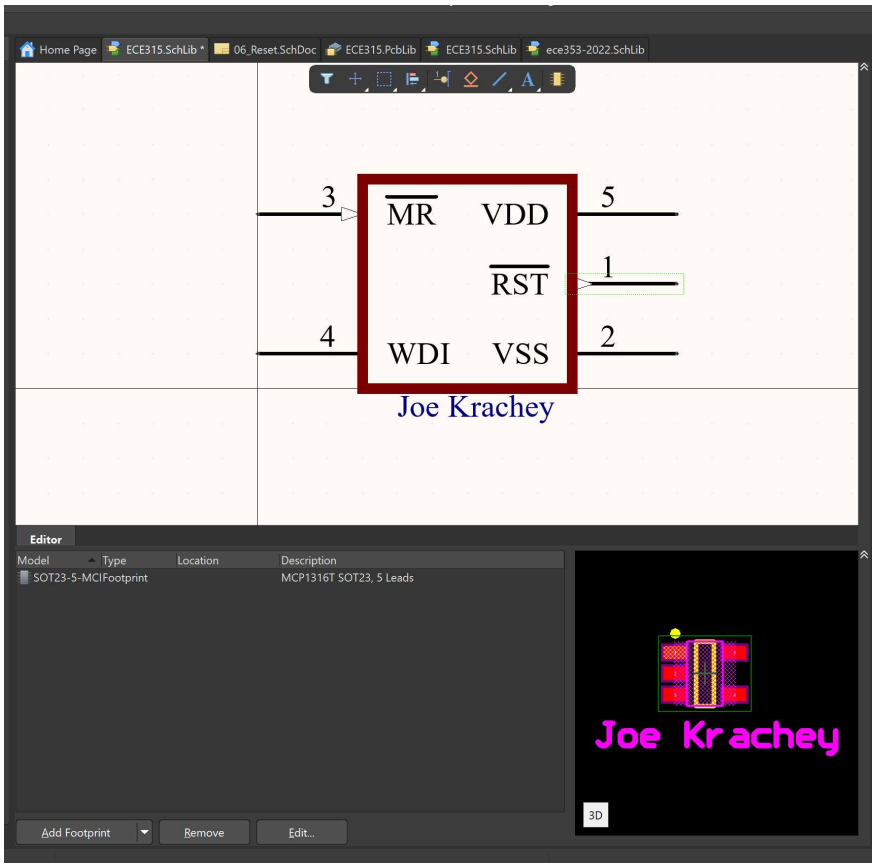


10. Schematic Symbols (MCP1316T-29LE/OT)

Create a schematic symbol for the MCP1316-29LE/OT.

Submission Requirement

1. Complete your Schematic Symbol for the MCP1316-29LE/OT
2. Use Snipping Tool to create an image of symbol along with the preview of the footprint
3. Save the file as Symbol-MCP1316-29.png
4. Upload the file to Lab 1 Assignment in Canvas.



11. Submit Altium Files

Submission Requirement

1. Create a Zip file that contains your Altium project folder.
2. Upload the Altium Zip file into Canvas.

12. Lab 1 Commentary

Creating all of the schematic symbols and PCB footprints might seem like a long and tedious process (it is), but luckily most companies employ a parts librarian that creates and maintains parts. This helps an embedded systems engineer to focus their efforts and develop products rather than reading mechanical drawings from datasheets.

If you're not lucky enough to work for a company that has a dedicated parts librarian, many parts have symbols and footprints that can be downloaded for free from sites like Digikey and Octopart. Starting in Lab 2, we will make use of Altium's Manufacturer Part's Vault for all of our additional parts. This will help to speed up the design process.